Practice Worksheet: Gerrymandering

Recall the 3 quantitative approaches to measuring gerrymandering discussed in lecture:
1. Mean - Median Score (Grofman, 1983)
2. Efficiency Gap (Stephanopolous and McGee, 2015)
3. Ensembles / Extreme Outliers

Use the following table to practice applying the first two techniques.

District	D Votes	R Votes	Result
1	62	38	D wins
2	55	45	D wins
3	44	56	R wins
4	47	53	R wins
5	46	54	R wins
Total	**254**	**246**	

Q1. Mean-Median Score

Calculate the mean-median score associated with this particular district breakdown of votes.

```r
# add code for Q1 here
votes <- votes |>
  mutate(d_share = d_votes / (d_votes + r_votes))

mean_median_score <- mean(votes$d_share) - median(votes$d_share)
mean_median_score
```

```
[1] 0.038
```

Q2. Efficiency Gap

Calculate the Efficiency Gap (EG) associated with this particular district breakdown of votes.

Tip: it may be helpful to write a function that takes a vector of votes across districts for party A and vector of votes for the party B and returns a vector of the number votes wasted by party A across districts.

```r
# add code for Q2 here
```